PROPERNESS OF SCHEME-THEORETIC SUPPORTS IS EVENTUAL

THE STACKS PROJECT AUTHORS

Editorial note. The source text concludes 'Z_i is proper over Y' where the statement asks over Y_i; the intended chain (X_i -> Y_i proper from the cited engine, then the closed support composes) is complete in span; the Y/Y_i slip is recorded verbatim here.

Allowed standard prerequisites (editorial).

- Scheme-theoretic support. Morphisms, Lemma morphisms-lemma-support-finite-type
- Image proper is proper. Morphisms, Lemma morphisms-lemma-image-proper-is-proper

Source reference. Original reference 081D: sources/stacks-chow/tags.txt, source lines 10213–10213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 056J: sources/stacks-chow/tags.txt, source lines 6712–6712. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03GN: sources/stacks-chow/tags.txt, source lines 4534–4534. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W5: sources/stacks-chow/tags.txt, source lines 2470–2470. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W3: sources/stacks-chow/tags.txt, source lines 2468–2468. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Assumptions and notation as in Situation 081D. Let $\mathcal{F}_0$ be a quasi-coherent $\mathcal{O}_{X_0}$-module. Denote $\mathcal{F}$ and $\mathcal{F}_i$ the pullbacks of $\mathcal{F}_0$ to X and X_i. Assume*

(1) *f_0 is locally of finite type,*
(2) *$\mathcal{F}_0$ is of finite type,*
(3) *the scheme theoretic support of $\mathcal{F}$ is proper over Y.*

Then the scheme theoretic support of $\mathcal{F}_i$ is proper over Y_i for some i.

Proof. We may replace X_0 by the scheme theoretic support of $\mathcal{F}_0$. By Morphisms, Lemma 056J this guarantees that X_i is the support of $\mathcal{F}_i$ and X is the support of $\mathcal{F}$. Then, if $Z \subset X$ denotes the scheme theoretic support of $\mathcal{F}$, we see that $Z \to X$ is a universal homeomorphism. We conclude that $X \to Y$ is proper as this is true

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

for $Z \to Y$ by assumption, see Morphisms, Lemma 03GN. By Lemma 081F we see that $X_i \to Y$ is proper for some i. Then it follows that the scheme theoretic support Z_i of $\mathcal{F}_i$ is proper over Y by Morphisms, Lemmas 01W5 and 01W3. $\square$